

JavaScript Promise Notes

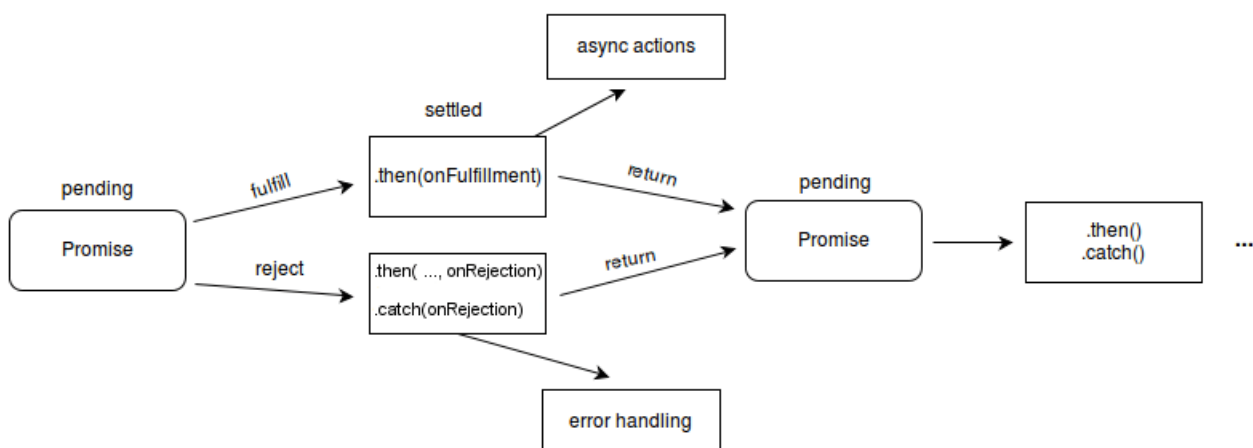
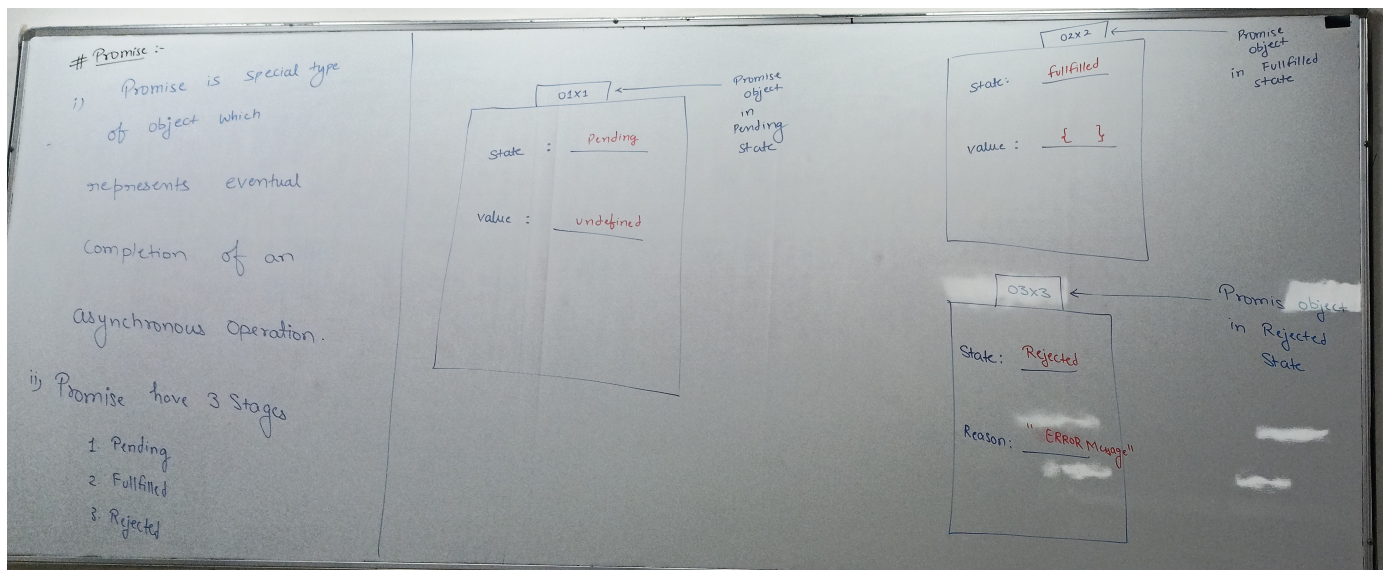
What is a Promise?

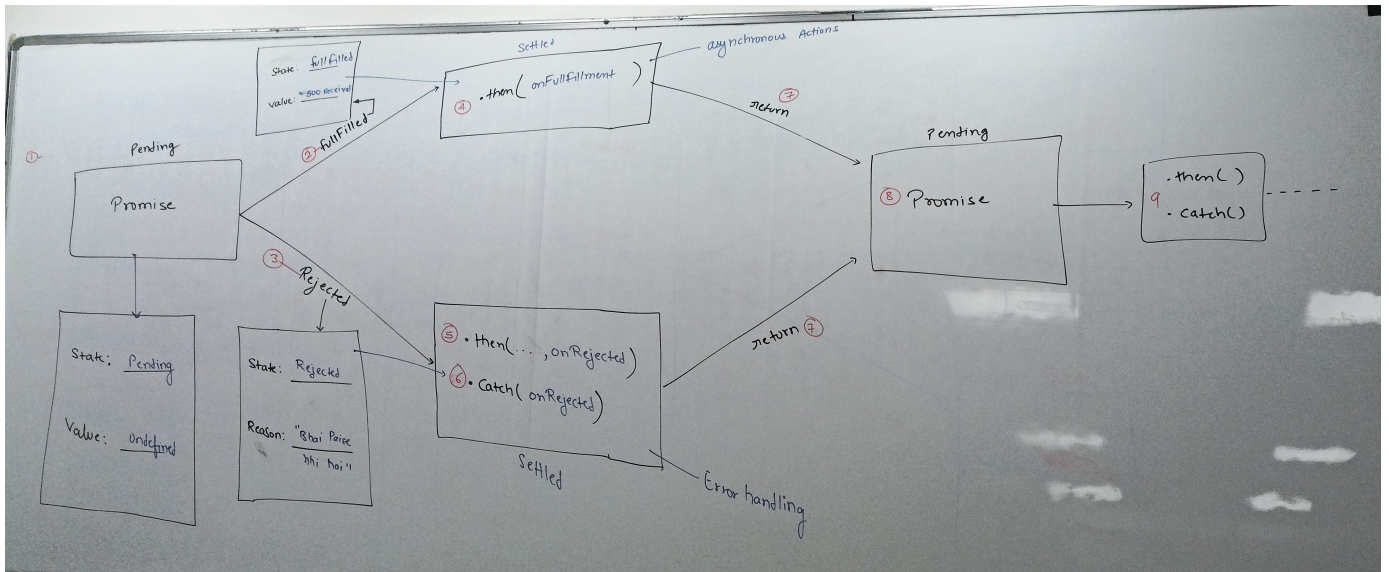
Promise is a special type of object which represents eventual (at the end) completion of an asynchronous operation (time taking operation).

Promise States

Promise has 3 stages:

1. **Pending** - Initial state, neither fulfilled nor rejected
2. **Fulfilled** - The operation was completed successfully
3. **Rejected** - The operation is rejected





Promise Constructor

Basic Syntax

```
const p1 = new Promise();
```

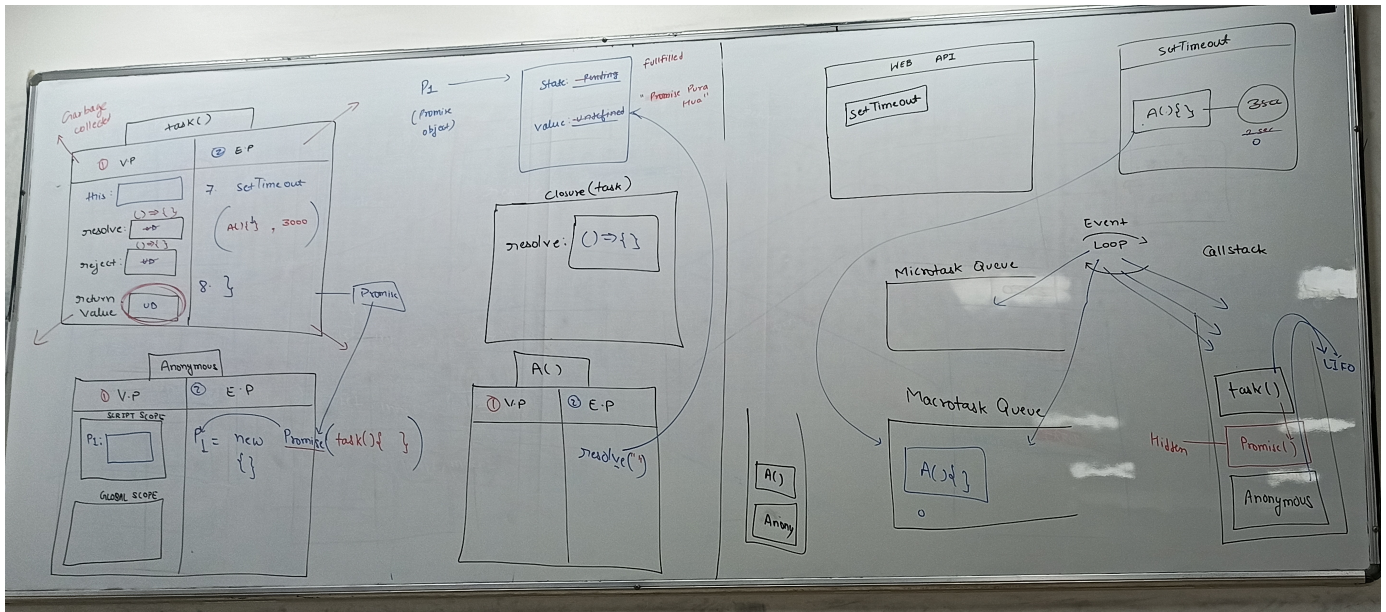
How it works:

- new** keyword loads the Promise() constructor to the p1 variable
- new** keyword creates an empty object: `{}`
- Empty object reference is saved to the Promise() constructor's **this** keyword

Promise Example

```
const p1 = new Promise(
  function task(resolve, reject) {
    setTimeout(() => {
      // reject("Yahi dosti yahi payr, bich me aa gai paise ki diwar");
      resolve("Five hundred received");
    }, 3000)
  });

console.log("p1 : ", p1);
```



Key Points:

- The `task` function is responsible for resolving or rejecting the promise
- Promise will be in **pending state** by default
- Use `resolve()` for successful completion
- Use `reject()` for failure/error cases

Additional Important Points

Promise Methods

- `.then()` - Handles fulfilled promises
- `.catch()` - Handles rejected promises
- `.finally()` - Executes regardless of promise outcome

Promise Chaining

```
promise
  .then(result => { /* handle success */ })
  .catch(error => { /* handle error */ })
  .finally(() => { /* cleanup code */ });
```

Promise Utilities

- `Promise.all()` - Waits for all promises to resolve
- `Promise.race()` - Returns first settled promise
- `Promise.resolve()` - Creates resolved promise
- `Promise.reject()` - Creates rejected promise

Common Use Cases

- API calls and HTTP requests
- File operations
- Database operations
- Timer operations (like setTimeout)
- Any asynchronous operation that takes time

Error Handling

Always handle promise rejections to avoid unhandled promise rejection warnings.

Promise() Constructor

```
const p1 = new Promise();
```

Explanation:

`new` keyword load the Promise() constructor to the p1 variable.

Note: new keyword create the empty object, like this: `{}`

empty object reference save ot the promise() constructor `this` keyword.

```
const p1 = new Promise(  
  function task(resolve, reject) {  
    setTimeout(() => {  
      resolve("Five hundred received");  
    }, 3000)  
  });
```

Here task function responsible for resolve or reject the promise.

Promise will in pending state by-default.

```
console.log("p1 : ", p1);
```

